# WACC/NPV RECAP

- **A business is planning to invest ZMW80m equity finance and ZMW40m debt finance is a 4-year project. The expected before tax net cash inflows of the project will be;**

| | | | | |
|---|---|---|---|---|
| **Year** | **1** | **2** | **3** | **4** |
| **Net cash inflows (ZMW'm)** | **50** | **80** | **70** | **40** |

- **Company tax is 25% per year payable in the same year as profits. *Capital allowances are available on a straight-line basis.* The project will have a nil scrap value at the end of 4 years. Cost of capital is 12.5%.**
- ***<u>Required:</u> Assess the viability of this project based on NPV***

# Capital allowances and tax shield

- **Capital Allowances = ZMW120m/4 = ZMW30m per year**
- **Tax shield = Capital allowance * tax rate**
- **Tax shield due to capital allowances (ZMW30m*0.25) = ZMW7.5m per year (yr 1 to 4)**

# NPV (ZMW'M) {discount factor = (1/(1+r)^n)}

| Year | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| Cash inflows | | 50.0 | 80.0 | 70.0 | 40.0 |
| Tax at 25% | | (12.5) | (20.0) | (17.5) | (10.0) |
| Tax shield(30*0.25) | | 7.5 | 7.5 | 7.5 | 7.5 |
| Initial cost (80+40) | (120.0) | | | | |
| Net cash flows | (120.0) | 45.0 | 67.5 | 60.0 | 37.5 |
| Discount at 12.5% | 1.000 | 0.889 | 0.790 | 0.702 | 0.624 |
| Present Values | (120.0) | 40.0 | 53.3 | 42.1 | 23.4 |
| NPV | 38.8 | | | | |
| Comment: | The project is viable since npv is positive. | | | | |

# Adjusted Present Value (APV)

- **The APV method is an approach that evaluates the project and the impact of the financing separately.**
- **Used if a new project has a different financial risk (debt to equity ratio) from the company, i.e., the overall capital structure of the company changes.**

# Adjusted Present Value (APV)

- ***APV consists of two different decisions:***
- **APV = Investment decision + Financing decision**
- **= Base case npv + PV of financing side effects.**
- **Value of a geared project = Value of an all-equity financed project + Present value of financing side effects.**
- ***Value of an all equity financed project = the base case NPV.***

# APV-Investment decision

- **The project is evaluated as though it were being undertaken by an all equity company with all financing side effects <u>ignored.</u>**
- ***Therefore:***
  - **Once the <u>base case NPV</u> is identified, the PV of the financing side effects is evaluated.**
  - **Use a beta that reflects just the business risk, i.e. <u>asset beta</u>.**
  - **Asset beta is used to get the <u>cost of equity</u> used in the base case NPV calculation.**

# APV-Financing decision

- ***Financing cash flows consist of:***
  - **Issue costs (of equity and or debt)**
  - **Tax reliefs (savings).**
- ***As all financing cash flows are low risk (mostly predictable) they are discounted at either:***
  - **The cost of debt or**
  - **The risk-free rate.**
- **PV of financing cash flows is then added to the base case NPV to find the APV.**

# Grossing up

- **Where issue costs of finance exist, there is need to gross up the amount required to get the full amount needed to be raised *(inclusive of the issue costs)*.**
- **<u>Practice Question:</u>**
- ***The finance required for a planned investment is ZMW2.4m (net of issue costs). Issue costs are 4%. And the finance raised will also have to cover the issue costs.***
- **What are the issue costs and what sum will need to be raised altogether?**

# Grossing up and issue costs

- **The ZMW2.4m is 96% of the amount to be raised:**
- **Therefore, (ZMW2.4m/0.96) = ZMW2,500,000 will be needed.**
- ***Issue costs are 4%***
- **4% × ZMW2,500,000 = ZMW100,000**
- ***Issue costs can be calculated in one stage as:***
- **KMW2.4m × 4/96 = ZMW100,000**
- ***Issue costs of equity are NOT tax deductible.***
- ***Issue costs of debt are tax deductible.***

# Calculating APV.

- ***The base case NPV is used as a starting point. The costs and benefits of the financing are then added to find a final adjusted present.***
- **1.Base case NPV** **X**
- **2. PV of the issue costs:**
  - **Equity** **(X)**
  - **Debt** **(X)**
- **3. PV of the interest tax shield** **X**
- **Adjusted present value** **X**
- **Decision rule: A project is viable if the APV is positive.**

# Practice question 1

- **Noble plc has a project with the following cash flows:**
- **Year 0 1 2 3 4**
- **Cash flows ZMW (10,000) 1,500 2,500 4,500 5,000.**
- **The ungeared cost of equity is 10%.**
- **<u>Required:</u>** ***Calculate the base case NPV and comment on the results***
- **If the firm use ZMW6,000 of 8% debt to finance the project and face a tax rate of 30%. The debt will be reduced by ZMW1,500 at the end of each year. Gross issue cost of debt is ZMW338.6.**

***What is the APV of the project?***

# Discount factors at 10%

| Year | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| Discount factors | $1/(1.1)^0$ | $1/(1.1)^1$ | $1/(1.1)^2$ | $1/(1.1)^3$ | $1/(1.1)^4$ |
| | 1.000 | 0.909 | 0.826 | 0.751 | 0.683 |

# Base case NPV

| Year | Cash flows ZMW | DF at 10% | Present value ZMW |
|---|---|---|---|
| 0 | (10,000) | 1.000 | (10,000) |
| 1 | 1,500 | 0.909 | 1,364 |
| 2 | 2,500 | 0.826 | 2,065 |
| 3 | 4,500 | 0.751 | 3,380 |
| 4 | 5,000 | 0.683 | 3,415 |
| | | Base case NPV | 224 |

# PV of interest tax shield

- **Calculate the *interest payable each period***
- **Apply the tax rate on the interest payable to get the interest tax shield.**
  - ***Interest tax shield = interest payable * tax rate***
- **Discount the interest tax shield using the cost of debt before tax (interest on debt)**
- **Gross issue cost of debt = ZMW338.6**
- **Tax = 30%**
- **Net of tax issue cost = ZMW338.6*(1-0.3) =ZMW237.02**

# Present value of interest tax shield (PVITS)

| Year | Loan at Start ZMW | Repmt ZMW | Interest at 8% ZMW | Interest tax shield at 30% | Df at 8% (1+r)^-n | PVITS ZMW |
|---|---|---|---|---|---|---|
| 1 | 6,000 | 1,500 | 480 | 144 | 0.926 | 133 |
| 2 | 4,500 | 1,500 | 360 | 108 | 0.857 | 93 |
| 3 | 3,000 | 1,500 | 240 | 72 | 0.794 | 57 |
| 4 | 1,500 | 1,500 | 120 | 36 | 0.735 | 26 |
| | | | | | Total | 309 |

# Calculating APV.

| | ZMW |
|---|---|
| Base case NPV | 224 |
| **PV of the issue costs:** | |
| Debt | (237) |
| PV of the Interest tax shield | 309 |
| Adjusted present value | 296 |

***Decision: Project is acceptable because the APV is positive.***

## Practice question 2

- **MS plc is evaluating the purchase of a new machine costing K450,000. The machine will have nil scrap value the end of its 4-year life. This investment is expected to be financed partly by equity while K300,000 of the initial cost will be financed by a 15% bank loan. The loan is repayable in equal instalments over 4 years. Interest is payable on a declining basis.**
- **The machine will produce after-tax net operating cash flows of K113,000 per year for 4 years. These cashflows are before adjusting for depreciation tax shield. The ungeared cost of equity is 8%. The company can claim depreciation allowances on a straight-line basis down to nil scrap. Corporate tax is at 20% payable in the same year as profits. The after-tax issue cost of finance is K12,000.**
- **Calculate the Adjusted Present Value (APV) of the project and comment on the results.**